INTERPRETABILITY & "BLACK BOX" PROBING

MARBERTv2 Arabic Fake News — What Is the Model Actually Doing?

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MODULE 01

THE PROBLEM

Right answer, wrong reason — the Clever Hans Effect

$ Why Interpretability Matters

—— The Danger of Shortcut Learning

▶ A model can hit 99% accuracy just because it learned: 'Reuters' = credible, '!!!' = fake

—— What we need to prove:

▶ The model is analyzing claims and semantics — not picking up on dataset artifacts

▶ If it flags any tweet mentioning 'Riyadh' as fake — it learned noise, not news

▶ # # Domain 4 is the X-ray of the model's brain

▶ → Model: MARBERTv2 · Dataset: 47,485 Arabic tweets

MODULE 02

KEY METRICS

What we use to look inside the model

$ The 3 Interpretability Lenses

METHOD

- ► Integrated Gradients (IG)
- ► SHAP Values
- ► Attention Entropy
- ► Ablation (Masking)

WHAT IT REVEALS

- ► Which tokens drove this specific prediction?
- ► Fair credit per word — gold standard for XAI
- ► Is the model reading the whole sentence or fixating?
- ► Remove top-3 words — does the prediction flip?

MODULE 03

MODEL PERFORMANCE

Numbers first — then we look at WHY they are this good

$ MARBERTv2 — Overall Results

Accuracy	→	98.92%	
Macro F1	→	0.9856	
ROC-AUC	→	0.9992	← near-perfect
MCC	→	0.9713	
Total Errors	→	511 out of 47,485	(0.8% wrong rate)

Dataset: 35,667 Fake tweets + 11,818 Real tweets

[SUCCESS] Strong baseline established

GRAPH 01

CONFUSION MATRIX

Where exactly do the 511 errors happen?

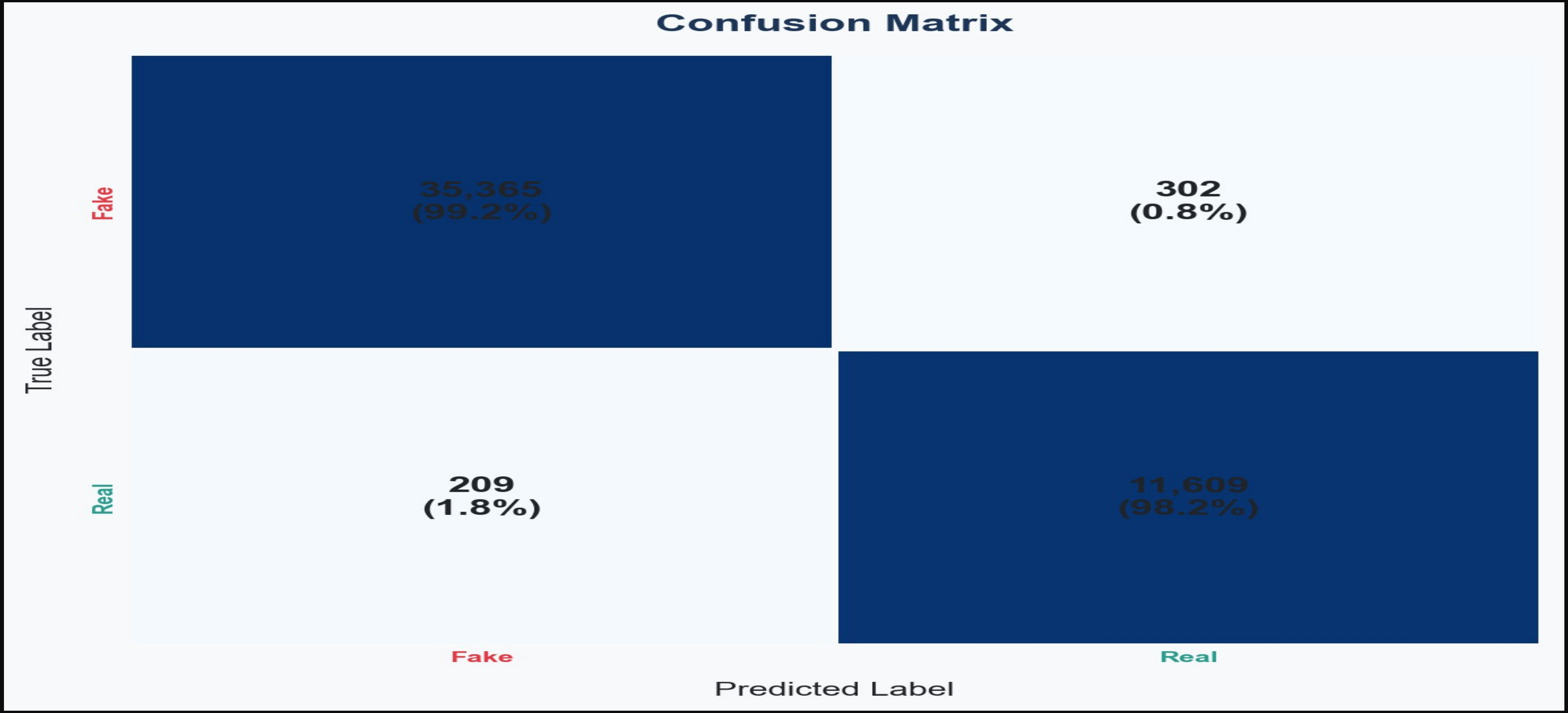
$ Confusion Matrix — The Numbers

True Positives	→	35,365	Fake correctly caught	(99.2%)
True Negatives	→	11,609	Real correctly passed	(98.2%)
False Negatives	→	302	Fake labeled as Real	← dangerous
False Positives	→	209	Real labeled as Fake	

False Negatives = misinformation that slipped through to readers

→ Strong diagonal dominance across both classes

Confusion Matrix — MARBERTv2 (47,485 samples)



TP=35,365 (99.2%) | TN=11,609 (98.2%) | FN=302 | FP=209 | Overall 98.92%

[IMG] marbertv2_confusion_matrix.png

GRAPH 02

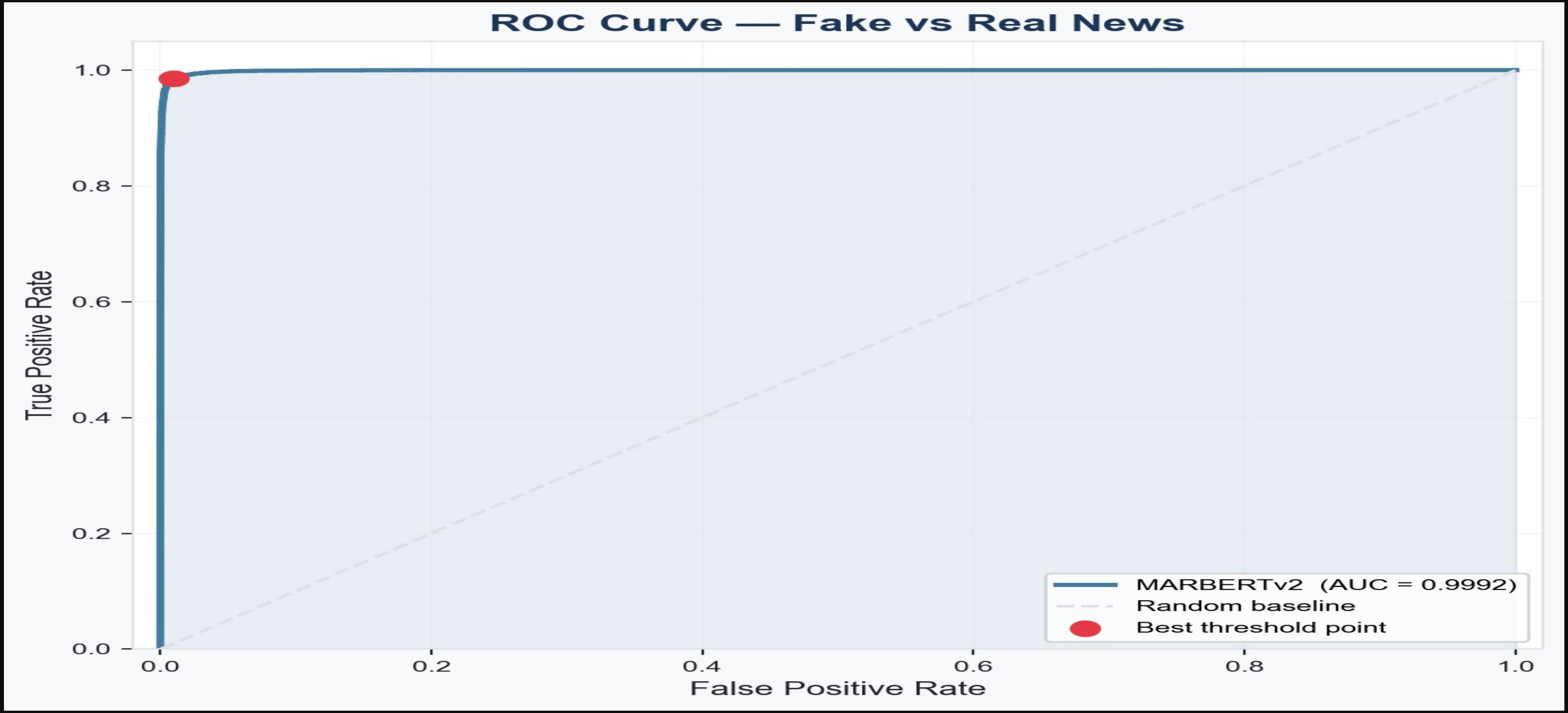
ROC CURVE — AUC = 0.9992

Near-perfect discrimination across all thresholds

$ ROC Curve — How to Read It

- Curve rises sharply to the top-left corner — ideal behavior
- AUC = 0.9992 → near-perfect separation of Fake from Real
- Red dot = optimal threshold: TPR ~98%, FPR ~1%
- Dashed diagonal = random classifier (AUC = 0.50) for comparison
- Compare: Abdullah's MARBERTv2 AUC was 0.565 — here it is 0.9992
- # # Confidence scores here are a reliable signal for thresholding

ROC Curve — Fake vs Real News



AUC = 0.9992 — curve hugs the top-left corner | Best threshold: TPR ~98%, FPR ~1%

GRAPH 03

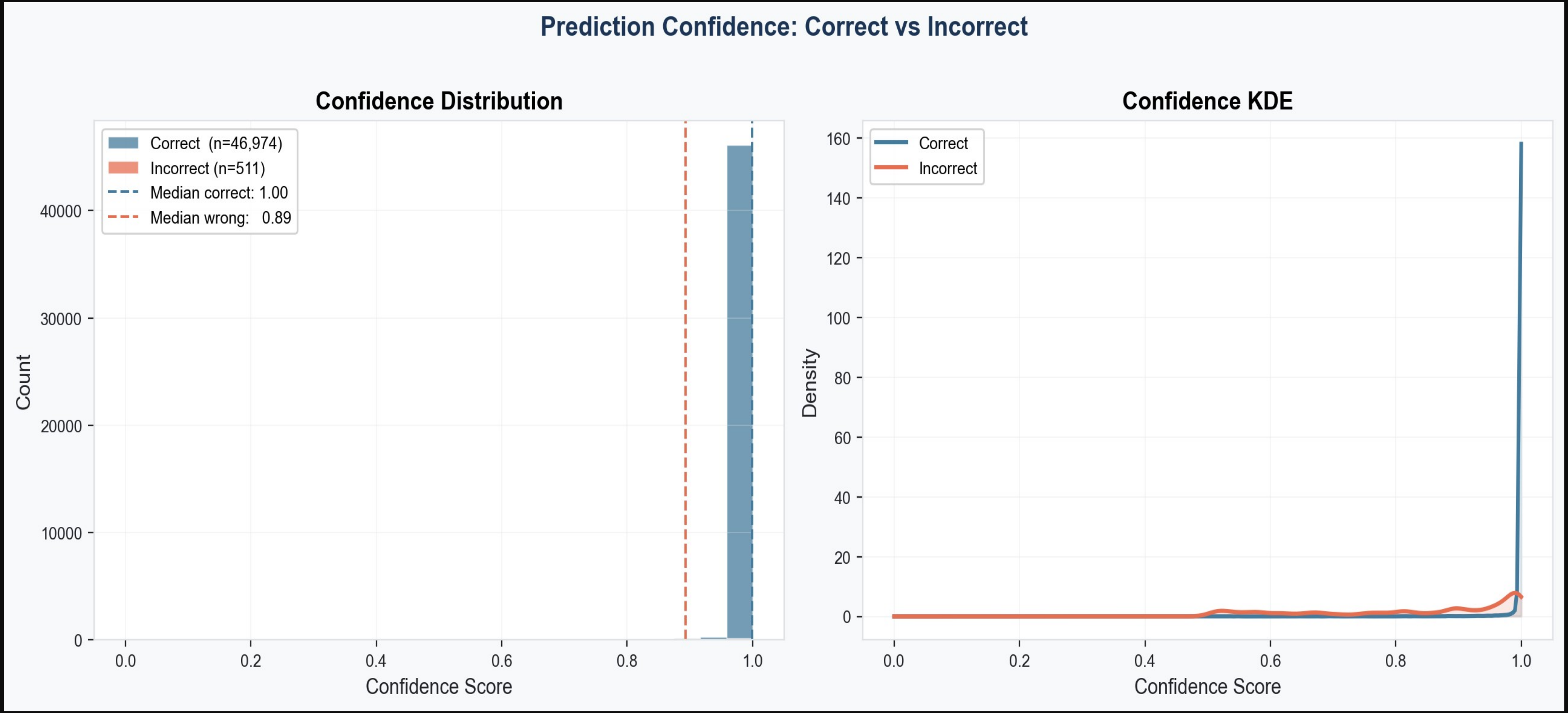
CONFIDENCE DISTRIBUTION

Does the model know when it's right vs wrong?

$ Prediction Confidence — Correct vs Incorrect

- ► Correct predictions (46,974): Median confidence = 1.00
- ► Incorrect predictions (511): Median confidence = 0.89
- ► Model is extremely certain when it's right — barely any spread
- ► When it's wrong, confidence is still high — but slightly lower than correct
- ► KDE plot: correct density spikes sharply at 1.0, incorrect spreads broadly
- ► # # This separation is a sign of a well-calibrated trustworthy classifier

Prediction Confidence: Correct vs Incorrect



Correct median = 1.00 | Incorrect median = 0.89 | Clear separation visible in both histogram and KDE

[IMG] marbertv2_confidence_dist.png

GRAPH 04

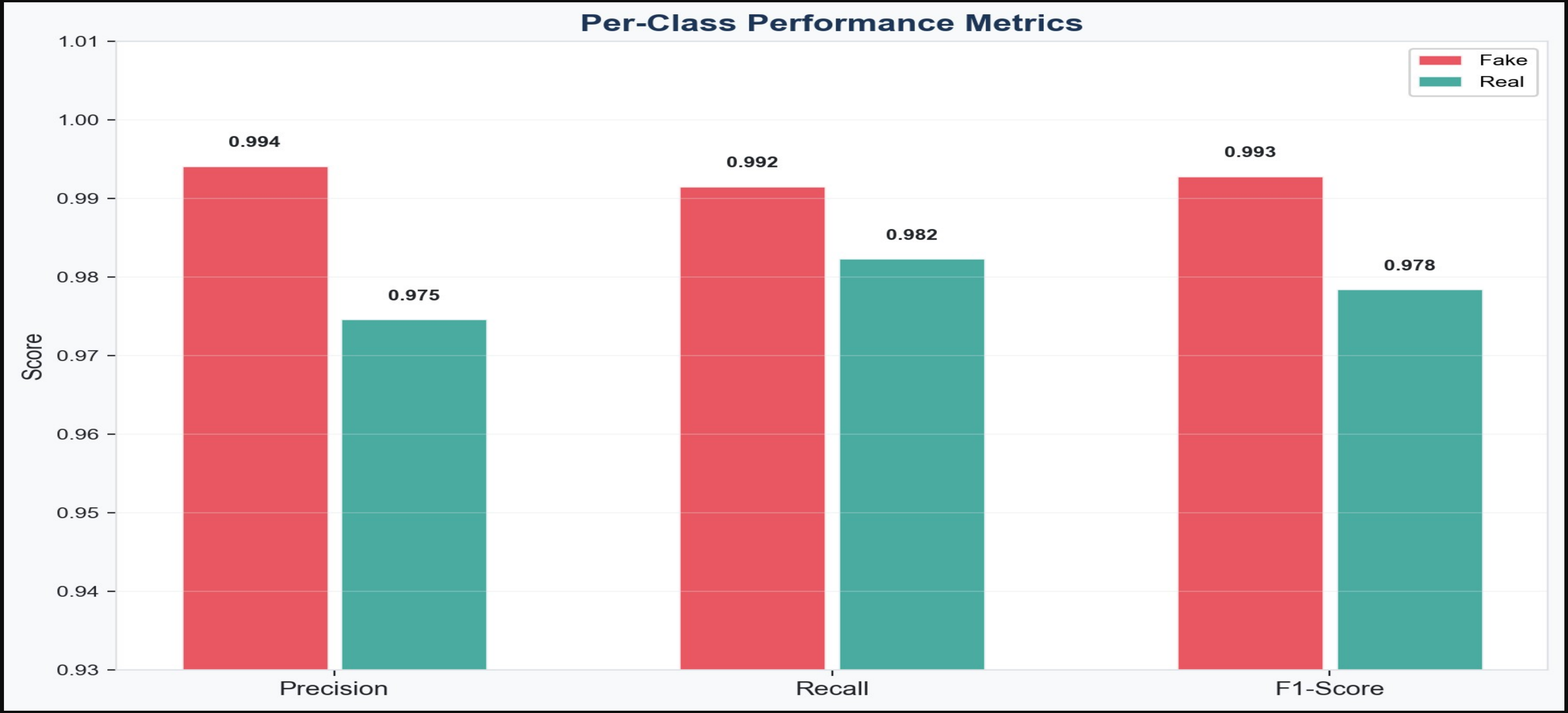
PER-CLASS METRICS

Does the model favor one class over the other?

$ Per-Class Precision / Recall / F1

- Fake class: Precision=0.994 | Recall=0.992 | F1=0.993
- Real class: Precision=0.975 | Recall=0.982 | F1=0.978
- Fake scores slightly higher — expected, it's the majority class (35,667 vs 11,818)
- Real class still achieves strong scores — model does NOT just predict the majority
- Macro F1 = 0.9856 — balanced across both classes
- # # No class bias — the model generalizes fairly to both Fake and Real

Per-Class Performance Metrics — Precision / Recall / F1



Fake: P=0.994 R=0.992 F1=0.993 | Real: P=0.975 R=0.982 F1=0.978 | Macro F1=0.9856

GRAPH 05

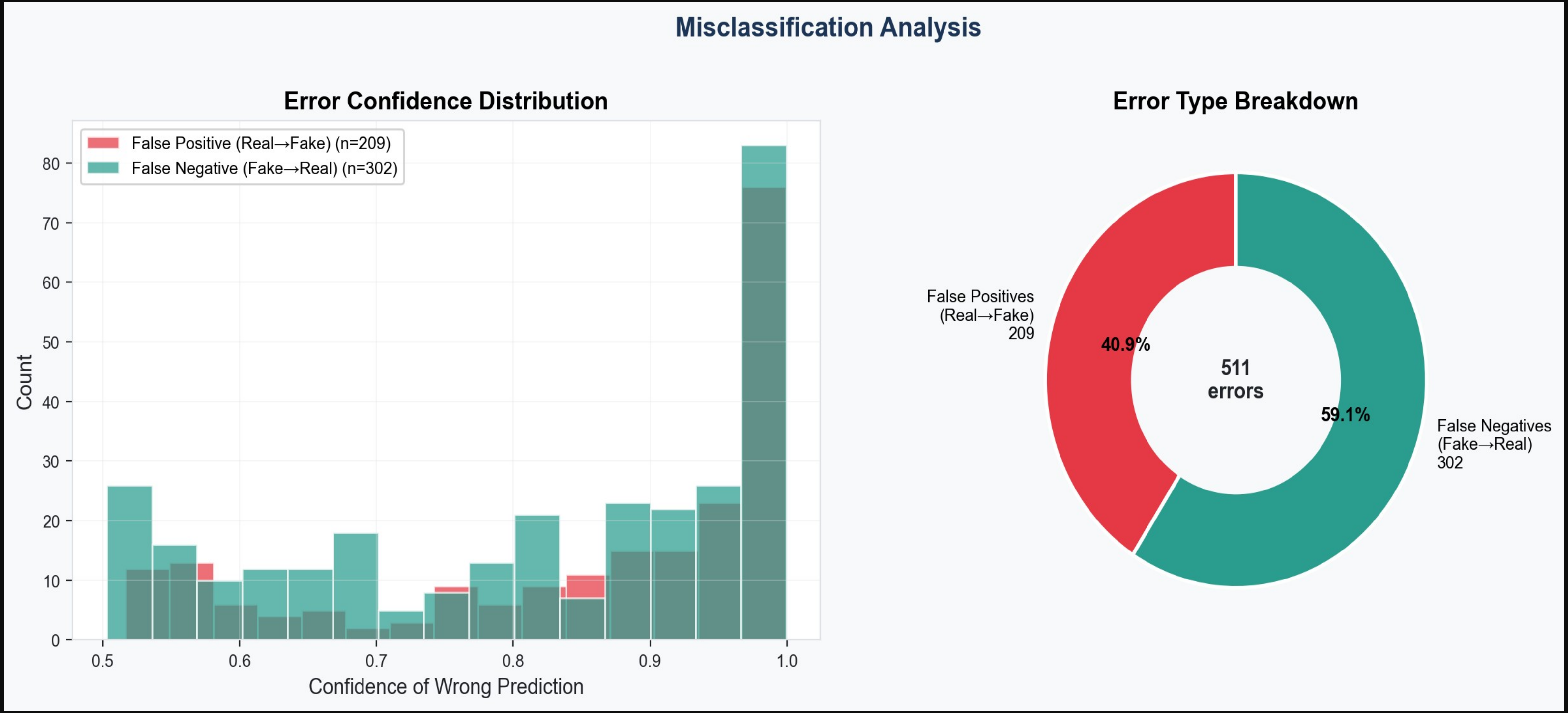
MISCLASSIFICATION ANALYSIS

What do the 511 errors look like — and when do they happen?

$ Error Analysis — The Dangerous Pattern

- False Negatives (Fake→Real): 302 (59.1%) ← more dangerous
- False Positives (Real→Fake): 209 (40.9%)
- Left histogram: many errors occur at HIGH confidence — model is wrong AND sure
- This is the overconfident mistake — system would never flag these for review
- Spike near confidence=1.0 in error histogram = the real hidden risk
- # # Ambiguous borderline tweets where surface language is misleading

Misclassification Analysis — Error Confidence + Type Breakdown



511 errors total: 302 FN (59.1%) + 209 FP (40.9%) | Most errors happen at high model confidence

[IMG] misclassification_analysis.png

GRAPH 06

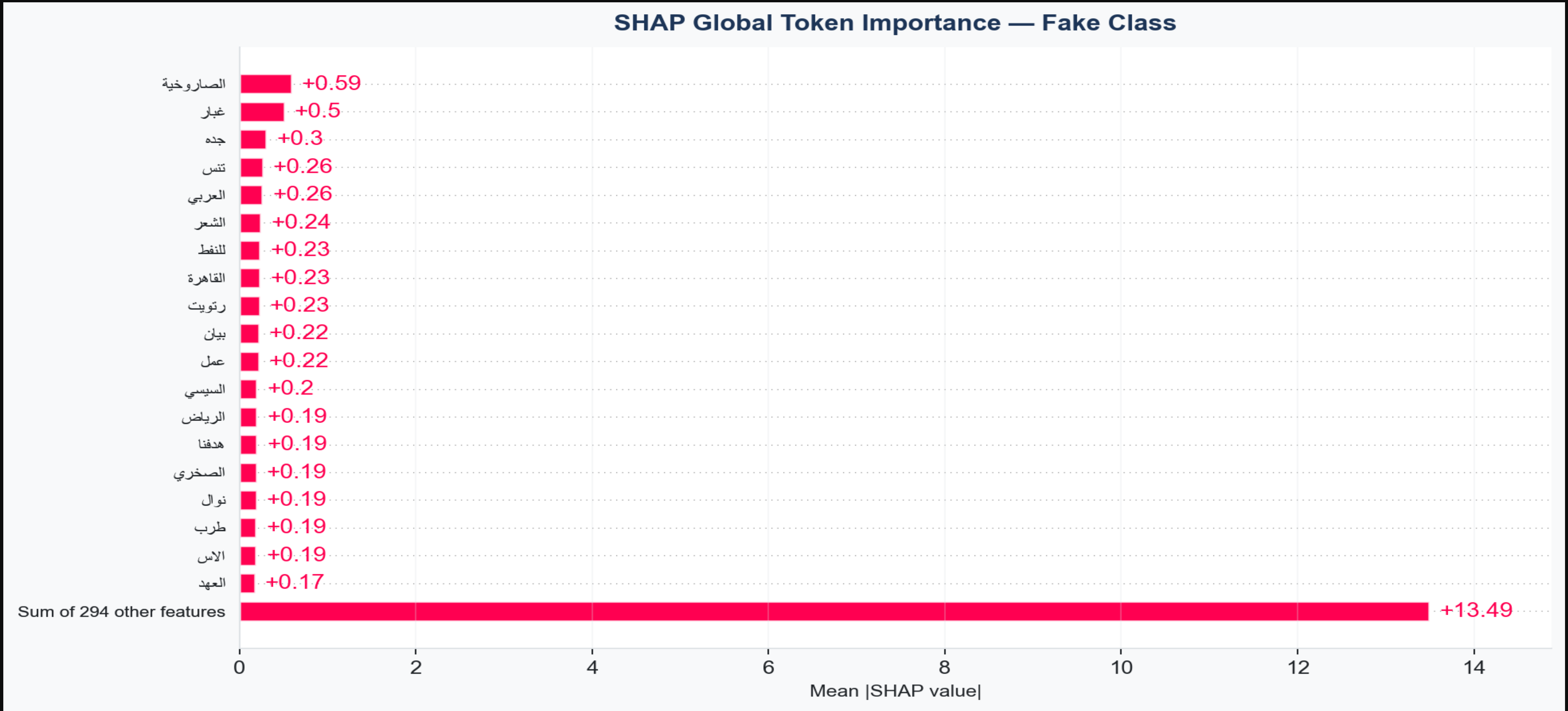
SHAP GLOBAL IMPORTANCE

Which words drive Fake predictions across the whole dataset?

$ SHAP Global Bar — Top Tokens for Fake Class

- ► Top token: الصاروخية (missile-related) → SHAP +0.59
- ► 2nd: غبار (dust) → +0.50 | 3rd: جده → 0.30+
- ► Named entities also appear: السيسي (0.20+), الرياض (0.19+), القاهرة (0.23+)
- ► These co-occur with fake content in this dataset — the model learned the pattern
- ► 294 other features collectively contribute +13.49 — no single token dominates
- ► # # Model uses hundreds of contextual signals — not a 'keyword detector'

SHAP Global Token Importance — Fake Class



Top: الصاروخية (+0.59) | Sum of 294 other features = +13.49 | Evidence spread across hundreds of tokens

[IMG] shap_global_bar.png

GRAPH 07

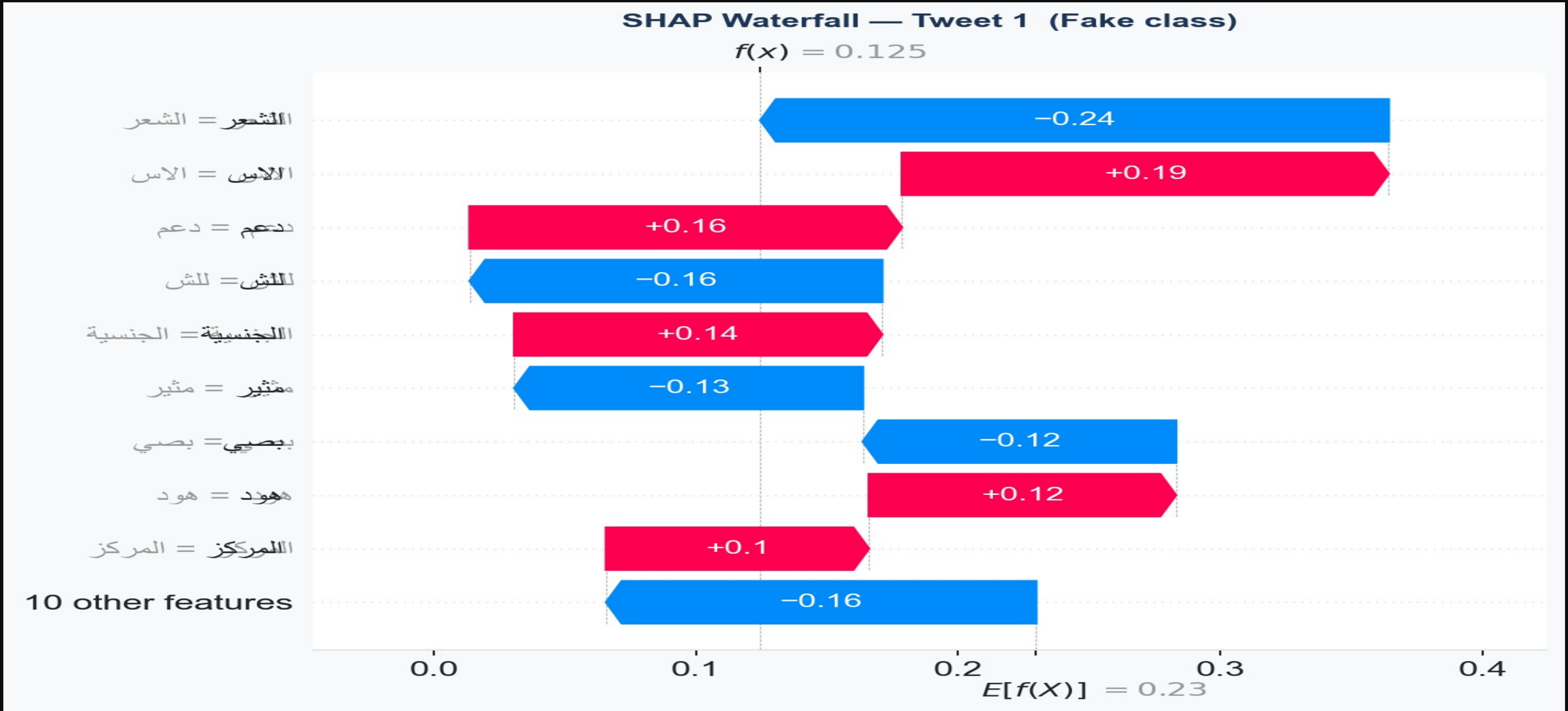
SHAP WATERFALL — TWEET 1

Step-by-step: why did the model classify this tweet as Real?

$ Waterfall Tweet 1 — Conflicting Signals

- ▶ Base value E[f(X)] = 0.23 | Final prediction f(x) = 0.125 → classified as Real
- ▶ الشعر (poetry/hair) → −0.24 ← strongest single signal, pushes toward Real
- ▶ دعم | 0.19+ → الاس (support) → +0.16 push toward Fake
- ▶ Conflicting signals cancel each other — final output below 0.5 threshold
- ▶ Even a correct prediction here relies on ambiguous features
- ▶ # # Model can be right for the wrong reason — this is what SHAP makes visible

SHAP Waterfall — Tweet 1 (Fake class probability)



Base=0.23 → Final=0.125 → Real | الشعر pushes −0.24 | Conflicting token signals balance each other

[IMG] shap_waterfall_1.png

GRAPH 08

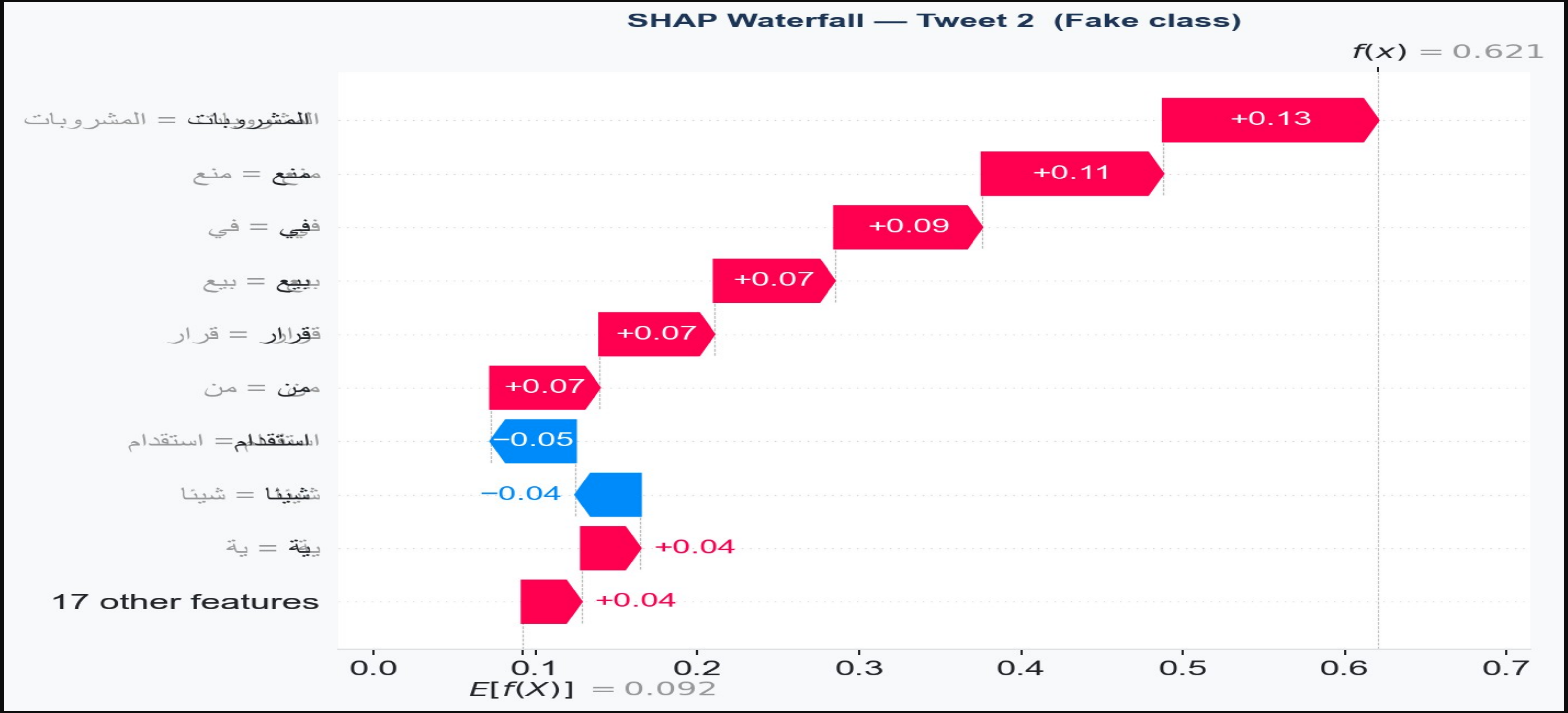
SHAP WATERFALL — TWEET 2

Coherent stacking — a well-justified Fake prediction

$ Waterfall Tweet 2 — Confident and Coherent

- Base value E[f(X)] = 0.092 | Final f(x) = 0.621 → classified as Fake
- المشروبات (beverages) +0.13 | ممنوع (banned) +0.11 | في (in) +0.09
- Nearly monotonically increasing — each token adds incremental Fake evidence
- Only 2 tokens oppose: استقدام (recruiting) −0.05 and شيئنا −0.04
- This is a high-confidence, well-justified prediction with coherent reasoning
- # # Many tokens consistently voting Fake = model reads the full topic, not one word

SHAP Waterfall — Tweet 2 (Fake class probability)



Base=0.092 → Final=0.621 → Fake | Monotonic positive stacking — coherent multi-token evidence

[IMG] shap_waterfall_2.png

GRAPH 09

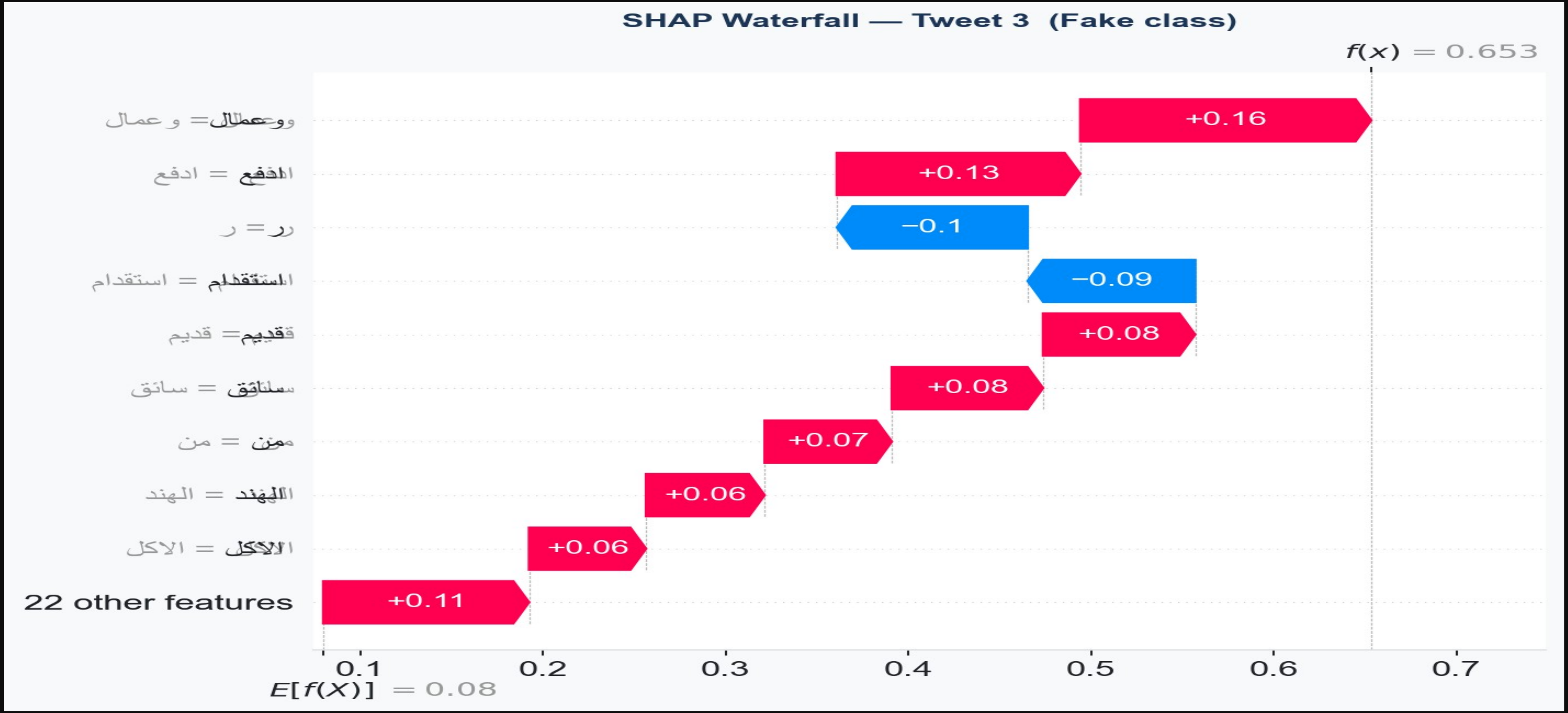
SHAP WATERFALL — TWEET 3

Economic keywords drive this Fake prediction

$ Waterfall Tweet 3 — Unique Feature Activation

- ► Base value E[f(X)] = 0.08 | Final f(x) = 0.653 → classified as Fake
- ► وعمال (workers) +0.16 | ادفع (pay) +0.13 | سائق (driver) +0.08
- ► Labor and economic theme dominates the positive signal
- ► Two tokens oppose: ر (sub-word fragment) −0.10 and استقدام −0.09
- ► Comparing all 3 waterfalls: each tweet activates a completely different feature set
- ► # # Model adapts reasoning per tweet — not a fixed fake-keyword vocabulary

SHAP Waterfall — Tweet 3 (Fake class probability)



Base=0.08 → Final=0.653 → Fake | Labor/economic keywords drive the signal | Each tweet = unique features

[IMG] shap_waterfall_3.png

GRAPH 10

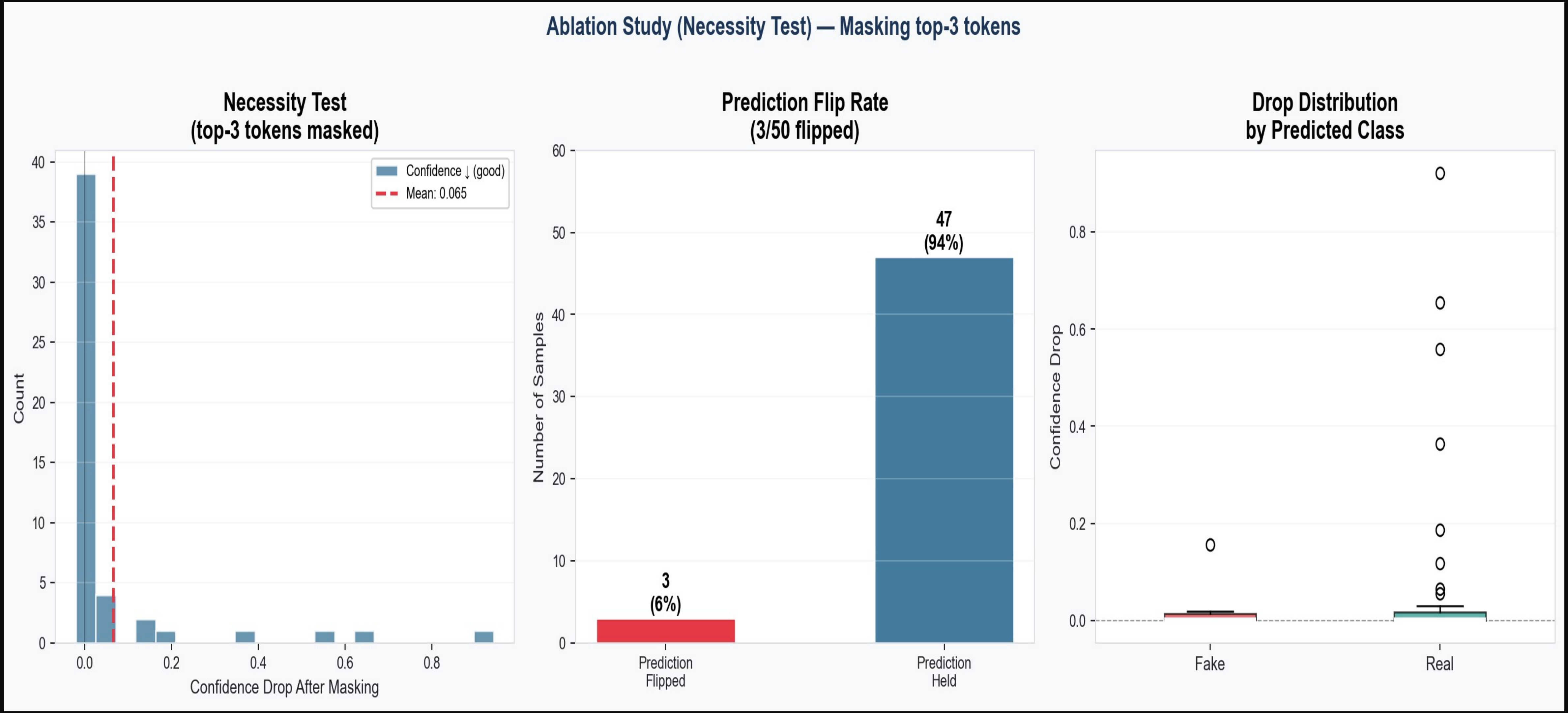
ABLATION STUDY

Mask the top-3 tokens — does the prediction flip?

$ Necessity Test — Top-3 Token Masking

- ► Masked the 3 highest-SHAP tokens in 50 tweets — measured confidence drop and label flip
- ► Mean confidence drop after masking: 0.065 — most samples barely moved
- ► Predictions flipped: only 3 out of 50 (6%) — model is robust to token removal
- ► Most samples cluster near zero drop — evidence spread across many tokens
- ► Real-class tweets show slightly more extreme outlier drops in the boxplot
- ► # # Not a shortcut learner — cannot be broken by removing one trigger word

Ablation Study — Necessity Test (Top-3 Token Masking)



Mean drop = 0.065 | Only 3/50 predictions flipped (6%) | Evidence distributed broadly

[IMG] ablation_conf_drop.png

GRAPH 11

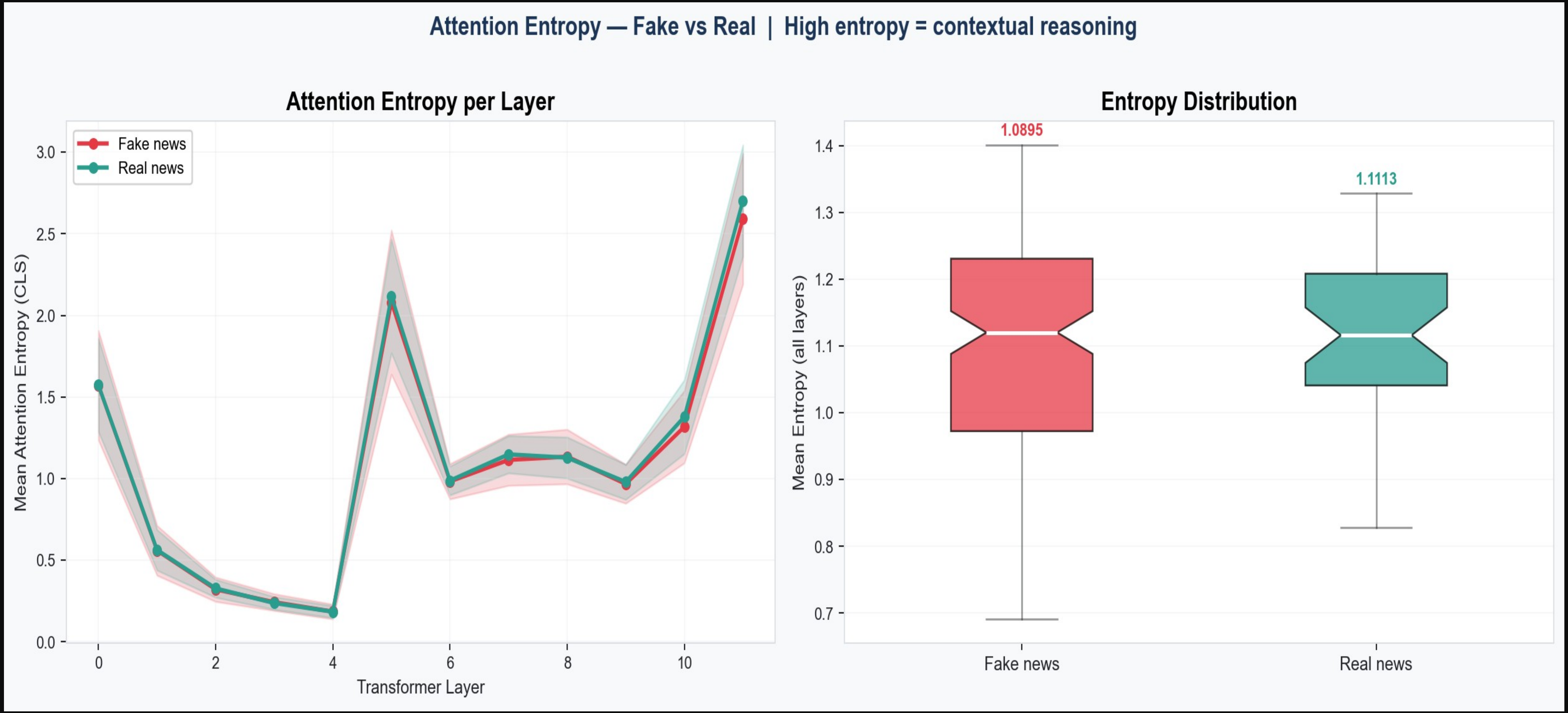
ATTENTION ANALYSIS

Is the model reading the full sentence or fixating on one word?

$ Attention Entropy + Head Visualization

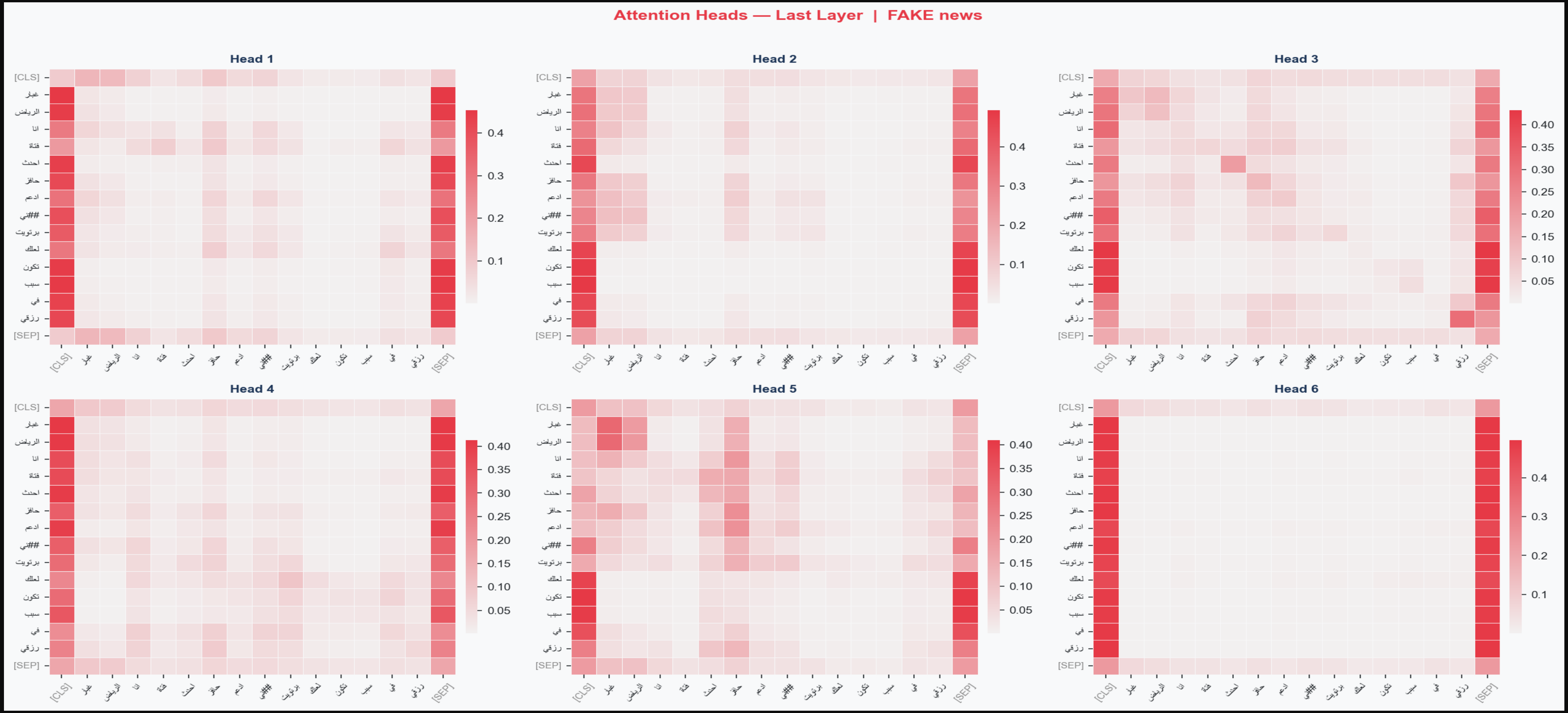
- ► High entropy = contextual reasoning | Low entropy = keyword fixation
- ► Fake news entropy: 1.0895 | Real news entropy: 1.1113 — both high
- ► Real news slightly higher — consistent with needing broader context to verify facts
- ► Fake news attention heads (red): Head 3 focuses on content words, Head 5 distributes broadly
- ► Real news attention heads (teal): Head 4 shows long-range dependencies across full tweet
- ► # # Both classes show contextual reasoning — not single-word trigger detection

Attention Entropy — Fake vs Real (Per Layer + Distribution)



Fake entropy=1.0895 | Real entropy=1.1113 | Both high — model reads full context for both classes

Attention Heads — Last Layer | FAKE News



Head 3: semantic focus on content words | Head 5: broad distributed pattern | Strong [SEP] attention expected

[IMG] attention_heads_fake.png

Attention Heads — Last Layer | REAL News

Head 4: long-range dependencies across full tweet | More distributed vs Fake heads — factual context reading

[IMG] attention_heads_real.png

CONCLUSION

FINAL VERDICT

Is MARBERTv2 actually reasoning — or just pattern matching?

$ Key Takeaways

── The model performs genuinely well

▶ 98.92% accuracy · AUC 0.9992 · Macro F1 0.9856 — strong across all metrics

── It is not a shortcut learner

▶ Ablation: only 6% flip rate after masking top-3 tokens — evidence spread broadly

▶ 3 waterfalls: each tweet activates a unique feature set — not a fixed keyword list

── One real risk remains: overconfident wrong predictions

▶ 511 errors — many at high confidence — model does not know it's wrong on these

▶ # # Fix: uncertainty-aware threshold + human review gate for high-confidence errors

[EOF]

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